Introduction to Trees

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Outline

1. Trees
2. Tree Terminology
3. Types of Trees
4. Properties of Trees
5. Tree Metrics
6. Summary

Introduction to Trees

Definition : Tree

A tree is a connected undirected graph with no simple circuits.

Key Properties

- Connected graph
- No cycles (acyclic)
- Minimal connectivity structure

Tree vs Non-Tree Examples

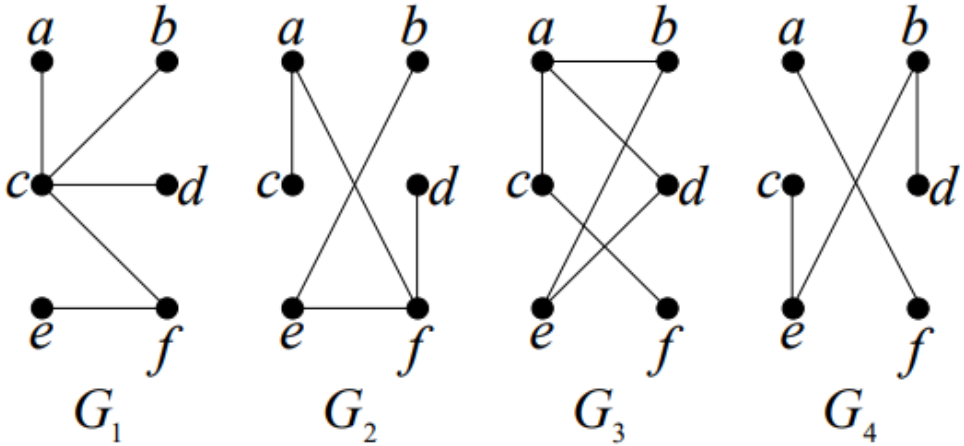


Figure: Some Example graphs

Observation

- G1 and G2 are trees
- G3 contains a cycle
- G4 is disconnected.

Tree Characterization

Theorem 1

An undirected graph is a tree if and only if there is a unique simple path between any two vertices.

Interpretation

- Exactly one path between nodes
- No redundancy in connections

Definition: Rooted Tree

A rooted tree is a tree where one vertex is designated as the root and edges are directed away from it.

Key Idea

- Root defines hierarchy
- Direction is implicit (top $\rightarrow$ bottom)

Rooted Tree Examples

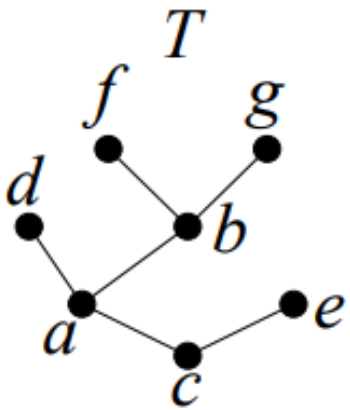


Figure: A Tree.

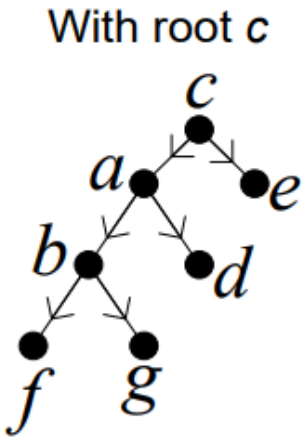


Figure: Rooted tree with c as root.

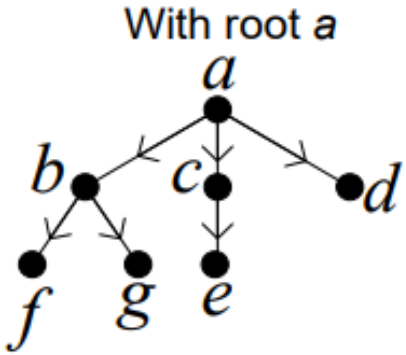


Figure: Rooted tree with a as root.

Basic Terminology

Relationships

- Parent: Immediate predecessor
- Child: Immediate successor
- Siblings: Same parent

Hierarchy

- Ancestors: Path from root
- Descendants: Subtree nodes

Definitions

- Leaf: Vertex with no children
- Internal Vertex: Vertex with at least one child

Subtree

A subtree rooted at vertex v includes: v and All its descendants

Tree Terminology Example

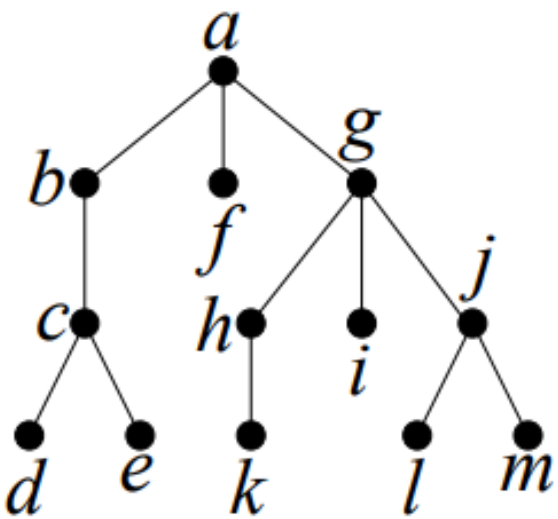


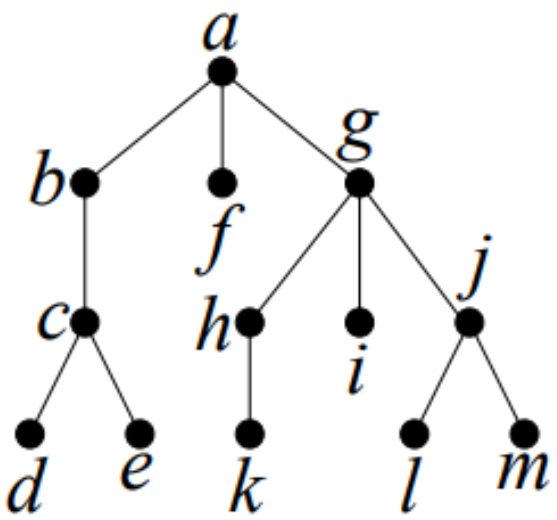
Figure: Example rooted tree illustrating parent, children, leaves, and subtrees.

Given Tree

Find:

- Parent of c
- Children of g
- Siblings of h
- Ancestors of e
- Descendants of b

Example Tree Analysis



Key Results

- Parent(c) = b
- Children(g) = h, i, j
- Siblings(h) = i, j
- Ancestors(e) = c, b, a
- Descendants(b) = c, d, e

m-ary Trees

Definition

- **m-ary tree:** A rooted tree where each internal vertex has at most m children.
- **Full m-ary tree:** Has exactly m children for every internal vertex.
- **Special Case:** For Binary tree, $m = 2$

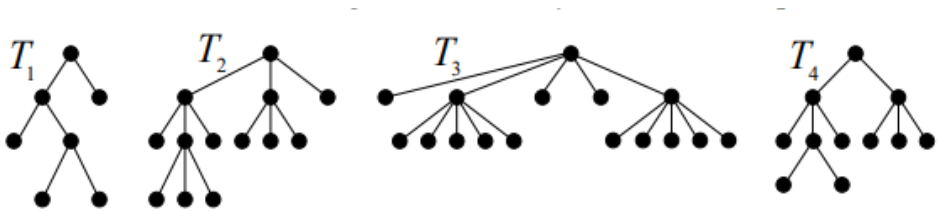


Figure: Examples of full and non-full m-ary trees.

Observation

- T_1 is full binary tree.
- T_2 is full 3-ary tree and T_3 is full 5-ary tree.
- T_4 is not a full m-ary tree for any m since some of its internal vertices have two children and others have three children.

Ordered Trees and Binary Trees

Ordered Rooted Tree

Children of each vertex are ordered (left to right).

Binary Tree

- Left child
- Right child

Subtrees

- Left subtree
- Right subtree

Binary Tree Example

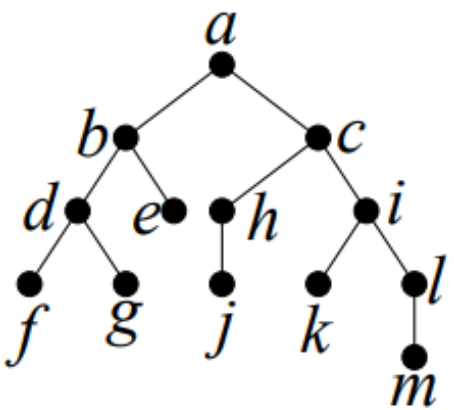


Figure: A Binary Tree T.

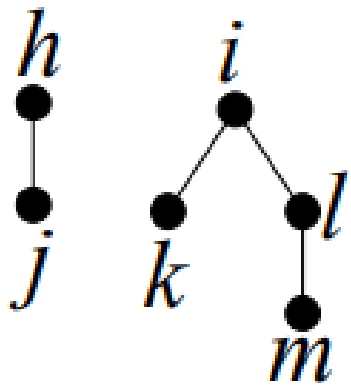


Figure: Left and Right subtrees of c.

Basic Tree Properties

Theorem 2

A tree with n vertices has $n-1$ edges.

Theorem: Full m-ary Trees

A full m-ary tree satisfies:

(i) If the tree has n vertices:

$$i = \frac{n-1}{m}, \quad l = \frac{(m-1)n+1}{m}$$

(ii) If the tree has i internal vertices:

$$n = mi + 1, \quad l = (m-1)i + 1$$

(iii) If the tree has l leaves:

$$n = \frac{ml-1}{m-1}, \quad i = \frac{l-1}{m-1}$$

Example: Tree Properties

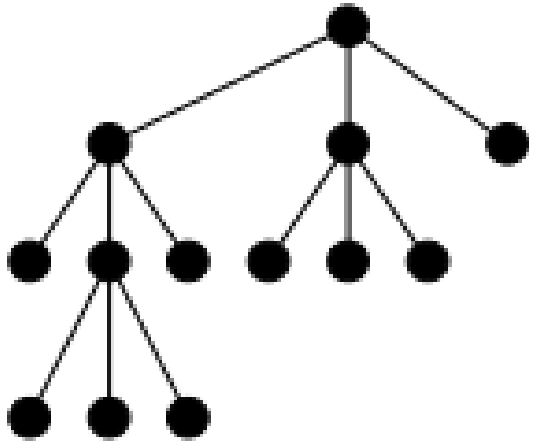


Figure: Example m-ary tree.

Given

Full 3-ary tree:

- $n = 13$

Results

- $i = 4$
- $l = 9$
- $e = 12$

Levels and Height

Definitions

- Level: Distance from root
- Root level = 0
- Height: Maximum level

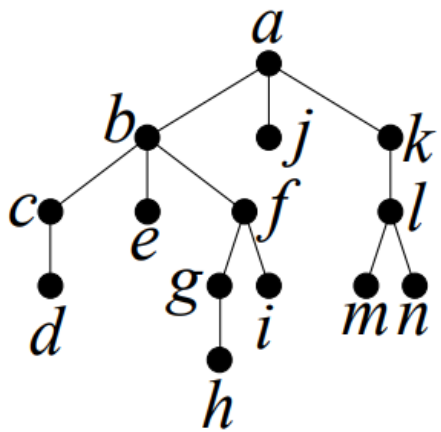


Figure: Rooted tree with vertices labeled to determine levels and height.

Problem

Find the level of each vertex in the rooted tree. What is the height of the tree?

Level Example

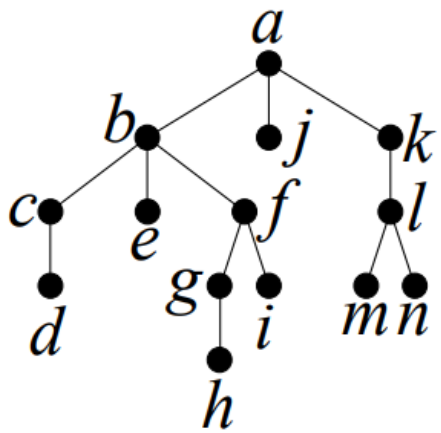


Figure: Rooted tree with vertices labeled to determine levels and height.

Solution

- Level 0: a
- Level 1: b, j, k
- Level 2: c, e, f, l
- Level 3: d, g, i, m, n
- Level 4: h

Height = 4

Balanced Trees Example

Definition

A rooted m-ary tree of height h is **balanced** if all leaves are at levels h or $h-1$.

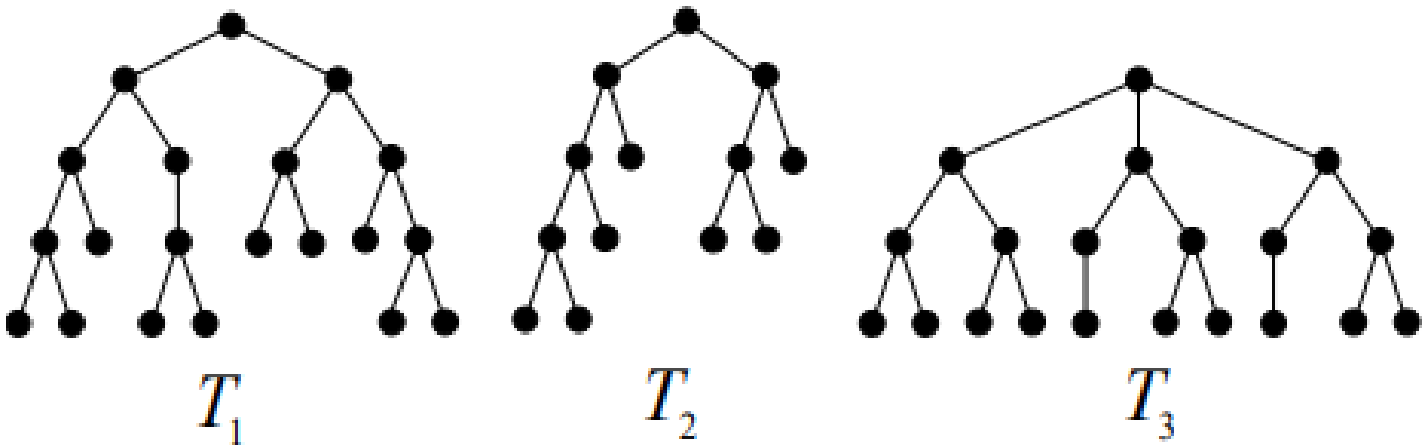


Figure: Some rooted Trees.

Problem

Which of the rooted trees T_1, T_2, T_3 are balanced?

Balanced Trees Example

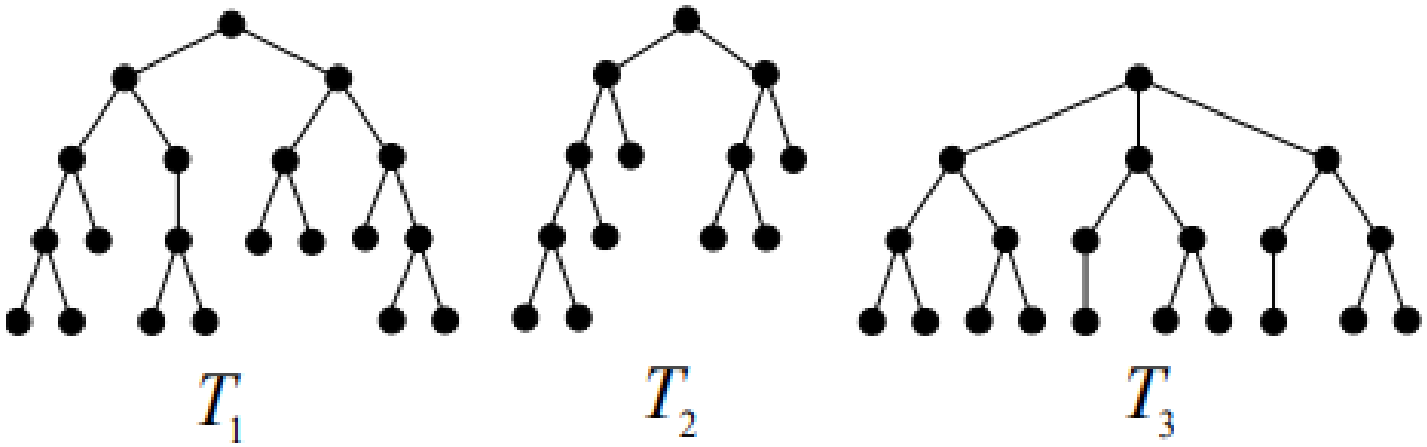


Figure: Some rooted Trees to be determined if they are balanced.

Solution

- T_1 is balanced (leaves at levels 3 and 4)
- T_2 is **not** balanced (leaves at levels 2, 3, and 4)
- T_3 is balanced (all leaves at level 3)

Summary

- Trees are connected acyclic graphs
- Rooted trees introduce hierarchy
- Tree terminology defines relationships
- m-ary and binary trees structure data
- Mathematical properties govern size and shape
- Levels and height measure depth

End of Lecture

Thank You